Unity Task1 : Introduction to Unity 6 with VR

1. Foundation & VR Mechanics

To begin the assignment, I completed a comprehensive one-hour introductory tutorial on Unity VR development. This provided a foundational understanding of the XR Interaction Toolkit and the VR Locomotion system. Specifically, I learned how to implement and configure core VR mechanics, including teleportation movement, XR Grab Interactables, and UI interaction using the XR Poke Interactor.

2. AI 3D Model Generation & Animation (Objective 2 & 3)

For the character asset, I utilized **Meshy.ai** to generate a 3D model of a medieval knight. During the initial generation phase, the AI produced a knight holding a sword in an irregular stance, which caused issues when attempting to rig the model. I learned that for automated rigging systems to work properly, humanoid figures must be generated in a clean, symmetrical "T-pose" or "A-pose" with empty hands.

After adjusting my prompt, I successfully generated a clean knight model and downloaded it as an `.obj` file along with its `.png` textures. I then imported this model into Adobe Mixamo, successfully mapped the skeleton, applied a "dying" animation, and exported the final animated character as an `.fbx` file for Unity.

3. Environment Assembly & Troubleshooting (Objective 4)

Building the surrounding world using Unity Asset Store environments presented several technical challenges, which I diagnosed and resolved:

- **Scale Discrepancies:** Some imported environments appeared microscopic or invisible. I learned that different modeling software uses different baseline metrics (centimeters vs. meters). I resolved this by adjusting the "Scale Factor" in the model's import settings.
- **Origin Stacking:** Assets occasionally loaded inside one another at the exact (0,0,0) coordinate. I learned to manually arrange these prefabs or utilize the creator's pre-arranged Demo scenes.
- **Material Compatibility (Pink/Gray Textures):** Older assets rendered without textures or appeared bright pink because their legacy materials were incompatible

with Unity 6's Universal Render Pipeline (URP). I utilized Unity's built-in Render Pipeline Converter to automatically upgrade these materials to URP.
- **Texture Atlases:** I also discovered an optimization technique used by 3D artists where a single "gradient" texture file serves as a color palette for multiple meshes. Re-linking this single texture to the base material instantly restored the colors to the entire low-poly scene.

4. VR Interactions & Physics Testing (Objective 1 & 5)

Finally, I populated the scene with medieval weapon assets and attached XR Grab Interactable components to them. During testing, I encountered a critical physics bug: the weapons immediately fell through the floor and tables. I deduced that while the Grab Interactable automatically adds a Rigidbody (which applies gravity), the objects lacked Colliders. Once I ensured that both the weapons and the environment surfaces (tables/floors) had the appropriate colliders (e.g., Convex Mesh Colliders), the physics engine functioned correctly.

Because I was developing without a physical VR headset, I implemented the **XR Device Simulator**. This allowed me to successfully enter the scene, navigate the environment, and test the Grab Interactables using keyboard and mouse inputs. I tested the three different grab Movement Types and observed how each handled physics and object clipping within the environment:

- **Instantaneous:** The object teleports directly to the hand's position every frame. While held, it completely ignores physics and static colliders, meaning it will clip cleanly through solid tables and walls.
- **Kinematic:** The object moves smoothly to the hand and acts as an unstoppable force. While it can push dynamic physics objects out of its way, it still ignores static colliders and will clip inside solid walls if pushed.
- **Velocity Tracking:** The most realistic setting, applying physical forces to move the object to the hand. It fully respects the environment's colliders; if you attempt to push a held object through a table, the table's Mesh Collider physically blocks it from passing through.

Video:
https://drive.google.com/file/d/1ESLU5yEebU4Gl8NAenyp-twRJw7l9buP/view?usp=drive_link